Environmental Data Science (2026), xx: 1–??
doi:10.1017/xxxx

CAMBRIDGE UNIVERSITY PRESS

METHODS PAPER

Deep Learning Ice Shelf Basal Melt Rates via Differentiable Physics

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Received: 23 January 2026; **Accepted:** 09 March 2026 (accepted for presentation at Climate Informatics 2026)

Keywords: Physics-informed machine learning, Fluid dynamics, Helmholtz decomposition, Antarctica, Ice shelves, Sea level rise

Abstract

Ocean-driven melting at the underside of Antarctica's floating ice shelves, a process known as basal melting, is responsible for over half of the continent's total contributions to global sea level rise. Despite its significance, basal melting is notoriously difficult to monitor due to the inaccessibility of sub-shelf cavities. Hence, models combine indirect remote sensing observations with mass balance physics to estimate basal mass balance, which represents the net effect of melting and refreezing at the base of the ice shelf. However, traditional methods frequently rely on fixed-grid regression, lower-order linear approximations, and simplified physical assumptions such as hydrostatic equilibrium, limiting their ability to resolve complex ice dynamics. We propose a novel physics-informed deep learning model that recovers continuous spatial fields of basal mass balance directly from sparse airborne ice shelf thickness and spaceborne ice velocity observations, while accounting for observable surface processes to ensure mass balance. Our proposed model, FluxNet, learns horizontal ice flux vector fields via a Helmholtz decomposition, and subsequently leverages automatic differentiation to derive the flux divergence field, thereby closing a critical gap in the mass balance equation. After validating FluxNet's ability to robustly reconstruct flux vector fields on holdout data, we provide a proof-of-concept by mapping basal mass balance across the Ross Ice Shelf, showcasing its utility as a physically interpretable, meshless, and efficient data-driven alternative to traditional methods.

Impact Statement

When warm ocean water flows underneath Antarctica's floating ice shelves, it melts the ice from below. This process, known as basal melting, is a main driver of Antarctic ice loss. Since direct monitoring of the ice shelf underside is unfeasible, capable models that integrate related observations and mass balance physics are needed to track this consequential melting process. We introduce a neural network framework — architecturally designed to model flows — for mapping ice flow divergence, which is the key unknown component in estimating basal melt. Our model addresses critical limitations of existing models, such as fixed grids and simplified physical assumptions, by learning continuous basal melt maps directly from observations. It thereby provides a promising practical, interpretable, and efficient alternative to existing methods.

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1. Introduction and Background

Although most of Antarctica's ice is grounded on land, its surrounding ice shelves, which are floating extensions of the land-based ice sheet, play a critical role in the stability of the entire system (Fricker et al., 2025; Fürst et al., 2016). These ice shelves provide a resisting force called *'buttressing'* that slows down the flow of grounded ice into the ocean, and thereby regulates the discharge of ice (Fürst et al., 2016). However, this stabilising quality is increasingly threatened by basal melting, the ocean-driven ablation of ice from below, which is estimated to account for more than half of the the continent's total mass loss (Rignot et al., 2013; Adusumilli et al., 2020). Covering an area comparable in size to Greenland, these ice shelves represent the critical interface between the ice sheet and the warming ocean, making the quantification of basal melt a high scientific and societal priority for constraining future sea level rise.

Directly monitoring the ice shelf underside remains physically unfeasible given existing observational possibilities (Fricker et al., 2025; Rignot et al., 2013). Consequently, basal mass balance (BMB) is typically treated as the unknown residual in the mass balance equation, inferred by reconciling indirect satellite observations of ice velocity, ice thickness, and elevation change with surface mass balance (SMB) modelled by regional climate models (Rignot et al., 2013; Depoorter et al., 2013; Adusumilli et al., 2020; Davison et al., 2023). While foundational studies established the first continent-wide temporal averages of BMB (Rignot et al., 2013; Depoorter et al., 2013), recent efforts have shifted towards high-resolution spatio-temporal characterisations (Adusumilli et al., 2020; Davison et al., 2023). These studies by Adusumilli et al. and Davison et al., which both utilise a similar Lagrangian mass balance framework but rely on different satellite data sources, represent the current benchmark for such spatio-temporal basal melt data products. Nonetheless, both operate on fixed grids and deploy linear plane-fitting approximations to derive ice shelf flux divergence, the primary dynamical term that represents

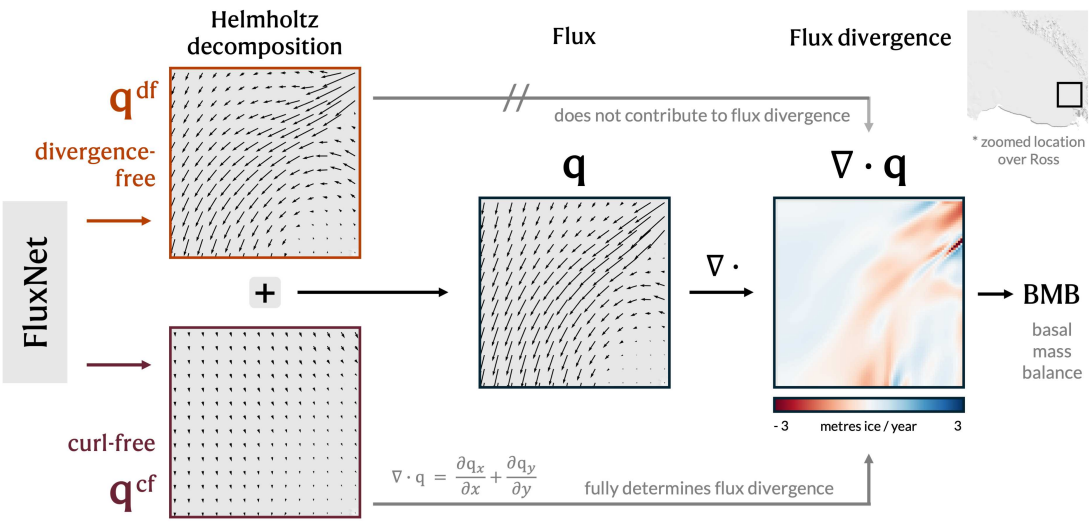

Figure 1. Helmholtz decomposition of ice shelf flux for a subregion of the Ross Ice Shelf: *FluxNet decomposes the flux vector field into its divergence-free component q^{df} and its curl-free component q^{cf}, which in sum recover the total flux q. The flux divergence $\nabla \cdot q$ is fully determined by q^{cf}, describing the spatial patterns of ice mass gain and loss. Note how the mass-conserving flux component q^{df} and the mass-redistributing component q^{cf} exhibit distinct magnitudes, directions, and spatial scales, that are not visually discernible from q alone. Here, the divergence-free transport q^{df} dominates the flux, while the divergence-driving dynamics q^{cf} occur almost perpendicular to q^{df}. By disentangling physically distinct transport and melt/freeze processes this decomposition improves interpretability.*

the horizontal redistribution of ice mass (see supplementary materials in Adusumilli et al., 2018). This reliance on discrete, lower-order approximations can smooths high-frequency spatial gradients, potentially obscuring localised melt features. Furthermore, rather than using ice shelf thickness survey data directly, both data products infer thickness by combining satellite altimetry, collected at dense spatial and temporal coverage, with assumptions of hydrostatic equilibrium. However, recent studies (Pritchard et al., 2025, see e.g. Fig. 4) indicate that this hydrostatic assumption introduces significant bias patterns in the inferred ice shelf thickness when compared to direct survey data.

This reliance on fixed-grid discretisations, linear approximations, and biased proxies for ice thickness highlight a critical need for meshless data-driven frameworks, capable of harmonising heterogeneous datasets while preserving high-resolution physical gradients. Machine learning (ML) has emerged as a practical tool to address such challenges in adjacent ice shelf modelling tasks, such as emulating computationally expensive ocean models to predict basal melt patterns (Rosier et al., 2023; Burgard et al., 2023), improving surface melt models (Hu et al., 2021), inferring distributions over complex physical parameters like ice rigidity through physics-informed variational deep learning (Riel and Minchew, 2023), and more recently, emulating and inverting ice shelf flow equations using physics-informed neural networks (PINNs; Raissi et al., 2019), a model that also leverages automatic differentiation (Wang et al., 2025). Building on these advances, we present a new solution for recovering ice shelf flux divergence fields, and to thereby monitor basal mass balance, grounded in physics-informed machine learning (PIML; Karniadakis et al., 2021). Our proposed framework, **FluxNet**,

- utilises a **mesh-free** representation that avoids resolution constraints and discretisation artefacts,
- computes **exact spatial gradients**, and hence divergence, via automatic differentiation,
- represents ice flux vector fields via a physically interpretable **Helmholtz decomposition**, (see Figure 1) that expresses flux as the sum of its two dynamically distinct flow components, and
- learns directly from non-uniformly distributed and **sparse survey data**.

We demonstrate the efficacy of our proposed approach through reconstruction experiments on ice flux data over the Ross Ice Shelf in Antarctica, evaluating performance on spatially held-out regions. We further generate a high-resolution basal melt map for the same domain, as a proof-of-concept application. Reproducible experiments, including data and PyTorch (Paszke et al., 2019) implementations are available at https://github.com/kimbente/FluxNet.

2. Method

To address the challenge of estimating basal mass balance (BMB) across Antarctica's floating ice shelves, we first formally define the problem and its underlying physical context, before introducing FluxNet, our proposed machine learning framework to model ice flux vector fields and their resulting flux divergence.

2.1. Problem Formulation

In this paper we address the problem of inferring basal mass balance (BMB) as modelling a spatial field over an ice shelf domain $\Omega \subset \mathbb{R}^2$. Formally, we model BMB as a function

$$(x, y) \in \Omega \subset \mathbb{R}^2 \;\mapsto\; \mathrm{BMB}(x, y) \in \mathbb{R} \tag{1}$$

where (x, y) denotes a horizontal location within the ice shelf domain Ω, and $\mathrm{BMB}(x, y)$ represents the local rate of mass exchange at the ice-ocean interface. We follow the sign convention where positive BMB values indicate basal freeze-on (mass gain) and negative values indicate basal melting (mass loss).

The physical basis for this inference lies in the principle of mass balance. As detailed by Morlighem et al. (2011), the Eulerian formulation describes the evolution of ice thickness h over time t for a given spatial location through the local mass budget, i.e., the balance between surface accumulation, basal processes, and ice dynamics. This is in contrast to the Lagrangian formulation used in Adusumilli et al.

(2020) and Davison et al. (2023), which tracks ice thickness along moving ice parcels. In this work, we adopt the Eulerian framework, in which this relationship is expressed by the *mass balance equation* (left):

$$\frac{\partial h}{\partial t} = \mathrm{SMB} + \mathrm{BMB} - \nabla \cdot (h\,\mathbf{v}) \quad \Longrightarrow \quad \mathrm{BMB} = \nabla \cdot (h\,\mathbf{v}) - \mathrm{SMB} + \frac{\partial h}{\partial t}. \tag{2}$$

Here, SMB is the surface mass balance, $\nabla\cdot$ denotes the divergence operator, and $\mathbf{v}$ is the depth-averaged horizontal ice velocity. By rearranging Equation (2) on the left, which yields the expression on the right, we can view BMB as the residual flux required to satisfy mass balance given the flow, surface effects, and changes in geometry.

In this study, we treat the surface mass balance (SMB) and the rate of thickness change ($\partial h/\partial t$, where positive values indicate thickening) as prescribed fields obtained from external observational and model-based datasets (see Figure 6 in the Appendix). Our primary modelling contribution focuses on the ice flux vector field, $\mathbf{q} = h\,\mathbf{v}$, and its resulting flux divergence, $\nabla \cdot \mathbf{q}$, i.e., $\nabla \cdot h\,\mathbf{v}$. In a 2D geospatial domain, the divergence of the flux is computed as the sum of the partial derivatives of its components in x and y directions,

$$\nabla \cdot \mathbf{q} = \frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} \tag{3}$$

where q_x and q_y are the respective components of the ice flux vector. Intuitively, $\nabla \cdot \mathbf{q}$ quantifies the local net outflow of ice flux: positive divergence indicates flux spreading (net export) from a location, resulting in dynamical thinning, while negative divergence indicates flux convergence (net inflow), leading to dynamical thickening.

2.2. FluxNet and Flux Divergence

To learn a continuous spatial representation of ice flux and to compute the associated flux divergence, we propose FluxNet. FluxNet is a machine learning framework situated at the intersection of physics-informed machine learning (PIML; Karniadakis et al., 2021) and geophysical modelling, which learns physically interpretable representations of flux vector fields by decomposing a field into its divergence-free and curl-free components via the Helmholtz decomposition shown in Figure 1 (see Bhatia et al., 2013 for theoretical fluid dynamics background). While FluxNet builds upon established neural architectures — namely, the Dissipative Hamiltonian Neural Network (DHNN; Sosanya and Greydanus, 2022) and model-inclusive learning (Kuroe et al., 1998) — we demonstrate how this formulation as an implicit neural representation (INR; Sitzmann et al., 2020) allows us to leverage automatic differentiation (autodiff; Baydin et al., 2018) to compute flux divergence exactly. By integrating this physically grounded inductive bias with modern differentiation tools, FluxNet enables the computationally efficient reconstruction of ice dynamics from sparse, irregularly distributed data (i.e., *interpolation*), yielding continuous flux divergence representations that transcend the resolution and mesh-based limitations of traditional discrete numerical approaches. The architecture of FluxNet and the subsequent computation of flux divergence are visualised in Figure 2. In essence, the model functions as an INR that maps coordinates to latent physical potentials, which are then differentiated to yield the ice flux vector field, and subsequently differentiated a second time to obtain the flux divergence, computed exactly via automatic differentiation. The specific steps of this process are as follows:

First, the network receives the spatial coordinates x and y as inputs, representing a location within the ice shelf geospatial domain. These inputs are then processed by a trainable neural network, which utilises a flexible and complexity-adjustable architectural backbone to map the coordinates to two latent representations of physical potentials. Specifically, this mapping yields a stream function $\psi(x, y) \in \mathbb{R}$ and a scalar potential $\phi(x, y) \in \mathbb{R}$, both of which are directly parameterised by the network weights. These potentials serve as the fundamental building blocks for reconstructing the ice flux field: By representing the flux through a stream function and a scalar potential, FluxNet adopts a formulation rooted in fluid dynamics known as the Helmholtz decomposition (see Bhatia et al., 2013; Kundu et al., 2016 for theoretical background). This fundamental theorem states that any sufficiently smooth vector

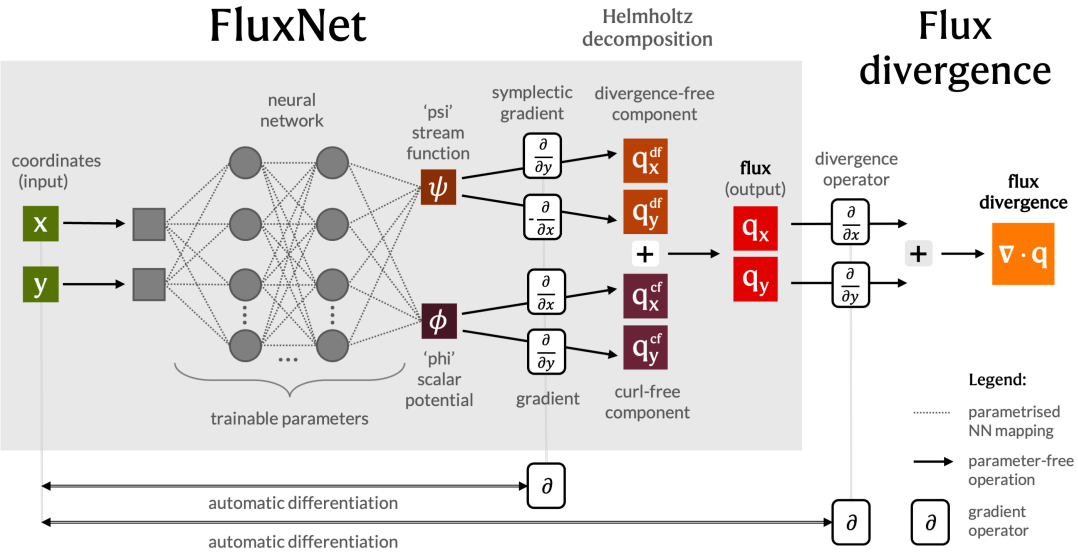

Figure 2. FluxNet architecture and divergence computation: *FluxNet first maps an input location to latent potentials ψ and ϕ. From these the divergence-free and curl-free vector field components, q^{df} and q^{cf}, which can be returned separately at inference time, are computed respectively using automatic differentiation. In sum both components represent the total flux q. A second automatic differentiation step yields the divergence. Note that only the neural network that outputs the potentials contains trainable parameters, while subsequent operations like differentiation and summation are parameter-free.*

field can be decomposed into a unique sum of a divergence-free (i.e., solenoidal) component $\mathbf{q}^{\mathrm{df}}$ and a curl-free (i.e., irrotational) component $\mathbf{q}^{\mathrm{cf}}$. To derive these flux components, the potentials are passed through a final, parameter-free layer that analytically maps ψ to the divergence-free component $\mathbf{q}^{\mathrm{df}}$ via the symplectic gradient ($\nabla^{\perp}\psi$), and ϕ to the curl-free component $\mathbf{q}^{\mathrm{cf}}$ via the gradient operator ($\nabla\phi$), both computed using automatic differentiation. Finally, FluxNet outputs the combined flux vector $\mathbf{q}$ as the sum of both vector field components, $\mathbf{q} = \mathbf{q}^{\mathrm{df}} + \mathbf{q}^{\mathrm{cf}}$. Since both flow components may often exhibit distinct spatial patterns, magnitudes, orientations, and spatial scales, the intuition is that their separate representations may also lead to a more nuanced and realistic composite representation. While this architecture does not strictly enforce a physical hard constraint as, for instance, the Hamiltonian Neural Network (HNN; Greydanus et al., 2019) — a related model class that only applies to the subset of flows that conserve mass or energy and hence only consist of the divergence-free component (also see Bente et al. (2025) who apply HNNs to ice sheets) — our model does impose a structural physical prior via the Helmholtz decomposition. Embedding this decomposition directly into the network architecture allows FluxNet to represent the full complexity of ice dynamics, enabling the model to capture the distinct smoothness and characteristics of each physical component, as illustrated in Figure 1, to aid generalisation, while isolating the divergent processes (q^{cf}) responsible for mass changes. Finally, to attain the flux divergence $\nabla \cdot \mathbf{q}$ as defined in Equation (3), the spatial gradients of the reconstructed flux components q_x and q_y are computed via a second round of automatic differentiation and summed. Because $\mathbf{q}^{\mathrm{df}}$ is divergence-free by construction, this process effectively extracts the Laplacian of the scalar potential ($\nabla^2\phi$) as the analytical representation of the flux divergence.

FluxNet (grey box in Figure 2) is trained end-to-end on ice flux observations using standard gradient-based optimisation. Importantly, only the neural network component has learnable parameters and subsequent gradient operations and summations (bold black arrows, Figure 2) are implemented as fixed analytical steps without parameters. This ensures that while the network adapts to fit the observational data, it is forced to do so through a physical framework that preserves the Helmholtz structure.

3. Experiments

As an initial evaluation, we apply our proposed FluxNet framework to **Ross Ice Self**, the largest ice shelf in Antarctica. Covering an area of approximately 500,000 km^2, roughly equivalent to the size of Spain, this region provides a real-world testing ground to evaluate FluxNet's feasibility and performance on vast spatial scales. Nonetheless, direct '*ground truth*' observations of BMB are essentially not available, making direct evaluation infeasible. We therefore assess FluxNet indirectly in two ways:

1. ***Flux Reconstruction Experiments***: First, since observations of ice flux are available, we spatially split these data into train and test regions to evaluate FluxNet's flux reconstruction performance against a standard 'physics-agnostic' neural architecture as well as a distance-based interpolation baseline.
2. ***Basal Mass Balance Mapping and Qualitative Comparison***: Second, training on all available flux observations over Ross Ice Shelf, we apply FluxNet to estimate flux divergence, and, after adjusting for SMB and thickening rates according to Equation (2), derive a BMB map as a proof-of-concept application. While a strict quantitative comparison with Adusumilli et al. (2020) is precluded by differences in the underlying datasets, we present both BMB maps side-by-side for a qualitative comparison.

3.1. Flux Reconstruction Experiments

Ice flux data. We combine ice thickness measurements h from the Bedmap3 source data collection (Frémand et al., 2023) — which largely originate from airborne surveys and are thus anisotropically distributed along flight lines — with ice velocity observations $\mathbf{v}$ from MEaSUREs phase-based ice velocity map (Mouginot et al., 2019). The 450 m resolution velocity grid is interpolated at the ice thickness coordinates to derive ice flux observations via the product $\mathbf{q} = h \cdot \mathbf{v}$, as shown in Figure 3.

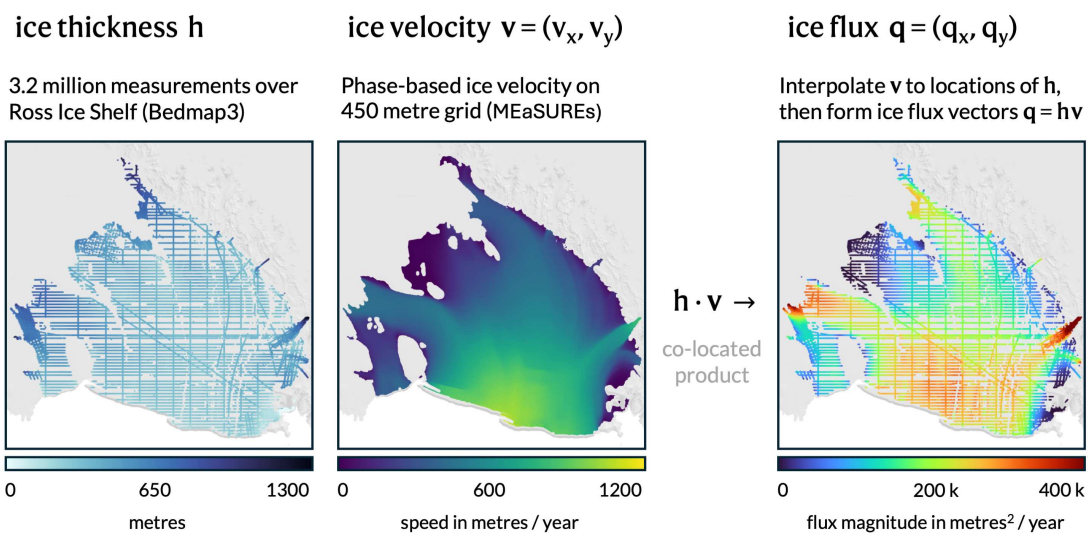

Figure 3. Ice flux dataset generation. *3.2 million ice thickness measurements h from the Bedmap3 source data collection (Frémand et al., 2023), distributed along flight lines over Ross Ice Shelf (left), are combined with dense satellite-derived ice surface velocity observations* $\mathbf{v}$ *from NASA's MEaSUREs phase-based Antarctica ice velocity map (Mouginot et al., 2019) (centre) into ice flux vectors* $\mathbf{q}$ *(right). By interpolating the dense* $\mathbf{v}$ *grid to the locations of h, and taking the product $h \cdot \mathbf{v}$ for each co-located scalar-vector pair, this yields a dataset containing ~3.2 million ice flux vectors. Here, ice flux is defined as the depth-integrated horizontal transport rate of ice mass, reported in metres^2 / year.*

Train-test split. To evaluate the model's ability to generalise across unsampled regions, we implement a spatial train-test split over the Ross Ice Shelf. The 3.2 million ice flux data points are partitioned into train (70%) and test (30%) sets using a 'gingham' pattern with 25 km wide strips, as shown in Figure 7 in the Appendix. This sampling strategy is specifically designed to emulate the linear geometry and spacing of realistic airborne flight patterns, allowing us to assess the model's capacity to interpolate between discrete geophysical survey tracks.

Baselines, Training, and Metrics. Performance is benchmarked by comparing our physics-informed FluxNet to both a a 'physics-agnostic' neural network model, as well as a distance-based interpolation baseline. Specifically, we use a residual Multi-Layer Perceptron (ResMLP) as the 'physics-agnostic' neural network baseline. While FluxNet derives flux from learned potentials via automatic differentiation, the ResMLP performs direct regression on the flux components. Both models share a common backbone of 6 residual layers and 256 hidden units, a configuration that we found to enhance training robustness while maintaining comparable and computationally tractable model capacity across architectures. For both machine learning models, we use the AdamW optimiser, a batch size of 1,024, a learning rate of 0.001, and train for 15 epochs. To account for stochasticity in the training procedure, we repeat all experiments across eight independent random initialisations. Further ablation studies, in particular on the choice of learning rate, are presented in the Appendix.

As a classical distance-based interpolation baseline we employ a k-nearest neighbours (KNN; Cover and Hart, 1967) method with inverse distance weighting (IDW; Shepard, 1968), KNN-IDW, in which predictions are computed as weighted averages of neighbouring observations, with weights decaying as a function of distance. While KNN-IDW is a deterministic algorithm with no trainable parameters, we investigate the sensitivity of the method to the choice of k, the number of nearest neighbours, evaluating $k \in \{1, 2, 4, 10, 16, 22, 28, 34\}$. Results are reported as the mean and standard deviation of the reconstruction MAE and RMSE across eight independent runs for ResMLP and FluxNet, or across the eight tested values of k for KNN-IDW. For further implementation details and to access the source code, please refer to https://github.com/kimbente/FluxNet.

3.2. Basal Mass Balance Mapping and Qualitative Comparison

To produce a high-resolution BMB map for Ross Ice Shelf, we train the FluxNet architecture on the full dataset of 3.2 million ice flux observations (see Figure 3). After training, we infer the flux divergence, $\nabla \cdot \mathbf{q}$, at a 500 m resolution, matching the Bedmap3 (Pritchard et al., 2025) grid, by calculating the divergence of the fitted flux field via automatic differentiation, as illustrated above in Figure 2. We then derive BMB by adjusting this divergence for surface mass balance (SMB) and thickening rates ($\partial h/\partial t$) according to the mass balance equation in Equation (2), as shown in Figure 6 in the Appendix. For the SMB component, we take the 1994–2018 annual average from Noël et al. (2023), which statistically downscaled ERA5-forced RACMO2.3p2 (van Wessem et al., 2018) from a 27 km to a 2 km resolution. We subsequently reproject these data to match our 500 m grid using bilinear interpolation. Similarly, ice shelf thickening rates are derived from 1994–2018 satellite radar altimetry from Adusumilli et al. (2020), adopting their methodology for translating surface height changes into depth-integrated thickening rates via firn and hydrostatic corrections. As evident in Figure 6, the spatial coverage of $\partial h/\partial t$ is restricted to 81.5°S due to orbital limitations of satellite altimetry missions prior to the 2010 launch of the CryoSat-2. While the native resolution of this data is 10 km, we reproject and interpolate it to our 500 m grid to ensure spatial consistency across all mass balance components.

Finally, we present the decadal-scale BMB map produced by FluxNet alongside the sub-decadal benchmark map of Adusumilli et al. (2020) to enable an initial qualitative comparison. While the benchmark corresponds to the 2010–2018 period and our data sources span a longer time horizon, this comparison provides a preliminary assessment of the respective qualitative characteristics of established altimetry-based maps and our proposed flux-based reconstruction, as well as their agreement in large-scale, long-term spatial patterns.

4. Results

4.1. Results from Flux Reconstruction Experiments

The predictive performance for the ice flux reconstruction task is summarised in Table 1, with a corresponding visual comparison of the error distributions provided in Figure 4. Note that the errors (reported in $m^2\,yr^{-1}$) are relative to a mean flux magnitude of 172,000 $m^2\,yr^{-1}$ across the study domain. FluxNet consistently outperforms both the KNN-IDW baseline and the physics-agnostic ResMLP: Our proposed architecture achieves a 22.1% reduction in MAE and a 31.4% in RMSE compared to KNN-IDW, and a 8.2% reduction in MAE and a more substantial 11.7% reduction in RMSE relative to the ResMLP (see columns on far right in Table 1). The larger relative improvement in RMSE suggests that FluxNet is particularly effective at reducing high-magnitude errors, indicating improved performance in regions with more extreme flux behaviour. FluxNet also demonstrates significantly greater robustness: the standard deviation across eight independent initialisations is approximately 48% lower than that of the ResMLP for MAE, and 46% lower for RMSE. By isolating the architectural difference (i.e., the Helmholtz decomposition), these experiments demonstrate that FluxNet enhances both predictive accuracy and generalisation stability relative to a standard MLP design, albeit both ML models outperform classical distance-based interpolation. Here, we found that the choice of k, the number of nearest neighbours considered in the inverse distance weighting (IDW), had negligible impact on reconstruction performance within this experimental set-up, where data are dense along flight lines but sparse in between. Overall, these results highlight the effectiveness of incorporating physically grounded structure for improving both accuracy and robustness in spatial flux reconstruction.

Table 1. *Test set predictive performance comparison of FluxNet, ResMLP, and KNN-IDW for ice flux reconstruction over the Ross Ice Shelf. Metrics are indicated in units of ice flux, i.e. $m^2\,yr^{-1}$. Results represent the mean ± one standard deviation across 8 independent runs in the case of FluxNet and ResMLP, and 8 different hyperparameter values for k in the case of KNN-IDW. Bold values indicate the best performance (i.e., lowest error). The two rightmost columns quantify the relative reduction in error achieved by FluxNet compared to both baselines.*

				Improvement of FluxNet over	
Metric	FluxNet (*proposed*)	ResMLP	KNN-IDW	ResMLP	KNN-IDW
MAE ± std	**5,829 ± 421**	6,349 ± 812	7,484 ± 34	8.2%	22.1%
RMSE ± std	**10,079 ± 478**	11,418 ± 883	14,682 ± 81	11.7%	31.4%

4.2. Resulting Basal Mass Balance map for Ross Ice Shelf

The left panel in Figure 5 shows the long-term average, decadal-scale BMB map based on FluxNet, which involved adjusting for modelled surface accumulation and observed thickness changes as illustrated in Figure 6 in the Appendix and described in Section 3.2 above. We observe intensified melting concentrated near the grounding lines and along the deepest portions of the cavity (refer to the left panel in Figure 3), where the freezing point of the ice is significantly lowered by the increase in pressure at depth. Furthermore, FluxNet resolves strong mass loss along the ice shelf margins, corresponding to high-activity calving zones. In contrast, the model identifies broad zones of basal refreezing in the shallower central regions of Ross. The right panel displays the subdecadal-scale, 9-year-average BMB map from Adusumilli et al. (2020). Although the primary melt and freeze-on patterns are qualitatively similar, discrepancies, such as broader melt regions resolved by Adusumilli et al. (2020) in central Ross, intensified melt patterns in the lower right corner of the map, south of McMurdo station (USA) and Scott base (New Zealand), as well as across the Eastern flank of the ice shelf, exist. These discrepancies may either be attributed to an intensification in basal melt over time, leading to higher melt rates in

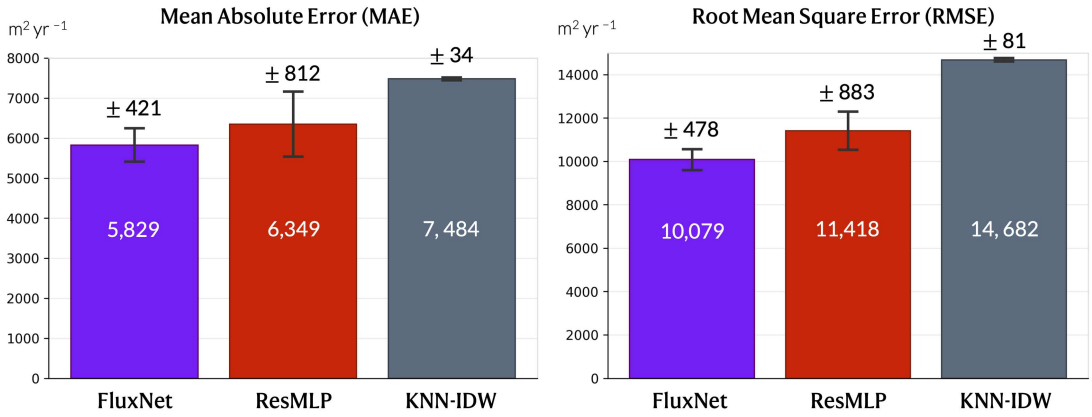

Figure 4. Bar chart test performance comparison, corresponding to Table 1: *Mean Absolute Error (MAE) for FluxNet, ResMLP, and KNN-IDW shown on left, where error bars represent* ± 1 *standard deviation calculated across all 8 experimental runs. Root Mean Squared Error (RMSE) comparison shown on right.*

the more recent 2010–2018 map on the right, or to differences in modelling approaches. In addition to these broader differences, and consistent with previously documented rift structures (e.g., see Figure 1 in Jiang et al., 2025), FluxNet's map more distinctly reveals the closing or migration of a former ice rift (dark blue strip) near the ice margin. Overall, the benchmark map exhibits pronounced satellite-related artefacts, whereas FluxNet yields a smoother and more spatially coherent reconstruction, free of such artefacts.

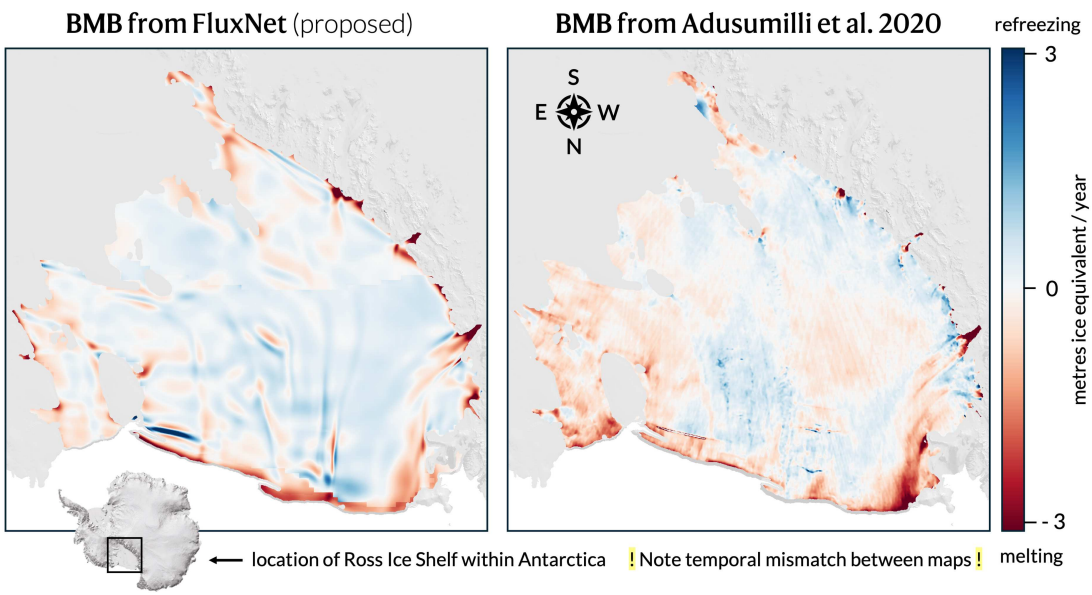

Figure 5. Spatial distribution of basal mass balance (BMB) across the Ross Ice Shelf*. Proposed BMB map based on FluxNet (left) versus the benchmark BMB map from Adusumilli et al. (2020) (right). Note that the benchmark corresponds to the 2010–2018 period, whereas our reconstruction integrates data over a longer time horizon. This temporal mismatch therefore limits the comparison.*

5. Discussion

Overall, the experiments confirm that FluxNet's physics-informed architecture provides an effective framework for flux reconstruction at scale. FluxNet's 11.7% reduction in RMSE compared to a physics-agnostic neural network baseline as well as the 31.4% reduction in RMSE compared to a distance-based interpolation baseline suggest it is particularly effective at mitigating outlier errors in high-flux regions. The performance gap between both neural models and the standard geospatial interpolation method indicates that INR models are better able to capture the underlying spatial structure and variability of ice flux fields than standard distance-based interpolation methods. Amongst the INR models, the lower standard deviation of FluxNet compared to ResMLP further highlights its robust and reliable performance. Building on these promising results on real-world flux data, the resulting BMB map (Figure 5) demonstrates FluxNet's ability to recover realistic mass balance patterns at fine spatial detail. Notably, we successfully mitigate satellite stripe artefacts inherent in Adusumilli et al., while producing broadly consistent melt features. Although in this initial study we use the complete historical record of ice thickness survey data, which precludes a strict pixel-by-pixel comparison of our temporal aggregate with Adusumilli et al. (2020) — which is only available for 2010-2018 — future studies shall integrate the latest generation of satellite altimetry data (e.g. by making temporally resolved adjustments of Bedmap3 ice thickness measurements using ICESat-2 or CryoSat-2 altimetry) together with annually resolved next-generation ice velocity products like *'ITS LIVE'* (Gardner et al., 2023). This will enable a more rigorous assessment of the framework's capability for spatio-temporal basal melt estimation and a deeper scientific investigation of the underlying physical controls on basal mass balance variability. While we assume that the lesser reliance of our proposed framework on the hydrostatic equilibrium assumption reduces the inherited biases (see ice thickness bias corrections over Ross Ice Shelf of up to 125 m in Figure 4 c in Pritchard et al., 2025), as discussed above, future work should systematically examine the implications of these differing conceptual approaches — namely, the use of direct ice thickness observations versus thickness inferred from satellite altimetry under hydrostatic equilibrium — on downstream estimates of flux divergence and basal mass balance.

Despite these limitations and open research questions, FluxNet's exceptional computational efficiency, training on 3.2 million data points in just 22 minutes on a single GPU, coupled with its ability to resolve accurate gradients, strongly motivates its application to such spatiotemporal, continent-wide use cases. While FluxNet provides an efficient deep learning framework for flux reconstruction, future research could incorporate probabilistic extensions, such as Bayesian hypernetworks (Krueger et al., 2018) or Helmholtz Gaussian Processes (Berlinghieri et al., 2023), to provide critical uncertainty quantification, or investigate whether a dual-branch structure, separating the neural network components that estimate the potentials ψ and ϕ, would provide additional performance increases.

6. Conclusion

FluxNet provides a robust, physically grounded, and interpretable framework for estimating basal mass balance by leveraging automatic differentiation to resolve flux divergence, the critical term in the mass balance equation. By significantly outperforming distance-based interpolation baselines, but also general physics-agnostic neural architectures in both accuracy and stability, the model demonstrates the potential of physics-informed deep learning as a scalable, mesh-free alternative for complex geophysical modelling. Our proposed approach successfully resolves fine-scale melt features while mitigating observational artefacts, offering an efficient path toward high-resolution monitoring of Antarctic ice shelf stability at a continental scale.

Acknowledgements. We thank Bryony Freer, Helen Fricker, Jason Roberts, Jakob Gradl, Anya Reading, and Felicity McCormack for insightful conversations during the development of this work.

Funding Statement. This work was partially conducted while KB was a PhD student at the University of Sydney, where she was supported by an Australian Government Research Training Program (RTP) Scholarship, as well as by the Australian Research Council (ARC) Industrial Transformation Training Centre in Data Analytics for Resources and Environments (DARE), project number IC190100031.

Competing Interests. None

Data Availability Statement. The code to reproduce the experiments in this paper is available at https://github.com/kimbente/FluxNet. All models are implemented in PyTorch. The experiments use the collection of geophysical measurements of Antarctic ice thickness from the Scientific Committee on Antarctic Research (SCAR) Bedmap project, the Bedmap3 source data collection, which was downloaded from the UK Polar Data Centre via the British Antarctic Survey (BAS) website (https://www.bas.ac.uk/project/bedmap/#data). The code repository contains scripts to programmatically download all standardised csv files from Bedmap1, Bedmap2, and Bedmap3 project iterations, and to preprocess the combined data. The experiments also require gridded ice velocity data. We use MEaSUREs Phase-Based Antarctica Ice Velocity Map, Version 1, which was downloaded from NASA's National Snow and Ice Data Center (NSICD) (https://nsidc.org/data/nsidc-0754/versions/1). Fully reproducible processing pipelines, as well as already preprocessed train and test data sets (`data/train_flux_tensor.pt` and `data/test_flux_tensor.pt`) are available in the github repository.

Ethical Standards. The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

Author Contributions. Contributions are listed in the order of the author list. Conceptualisation: K.B. Data curation: K.B. Methodology: K.B, R.M, F.R. Resources: F.R. Software: K.B. Supervision: R.M, F.R. Visualisation: K.B. Writing – original draft: K.B. Writing – review & editing: K.B, R.M, F.R. All authors approved the final submitted draft.

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Appendix

A. Basal mass balance composition

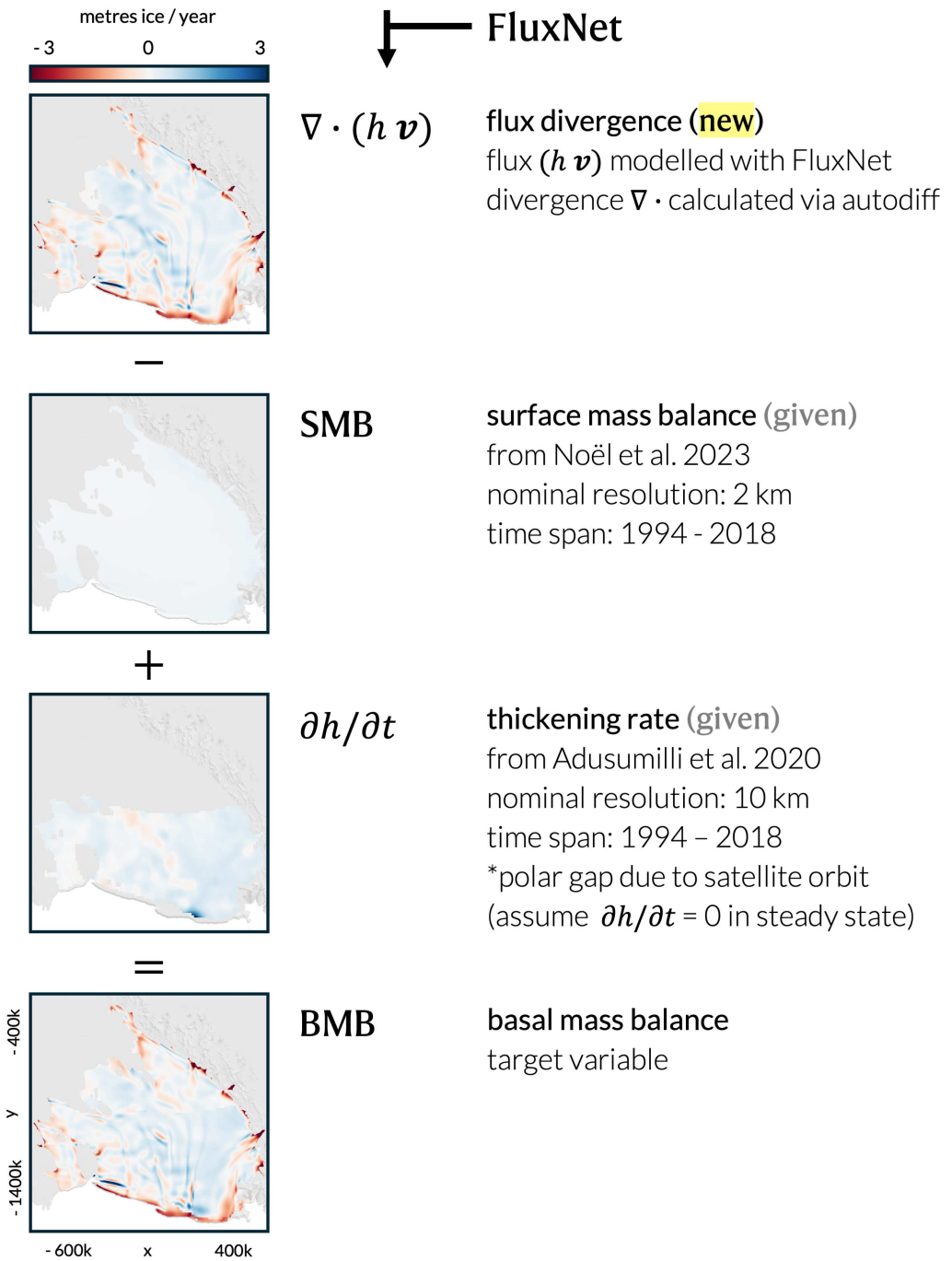

Figure 6. *Composition of basal mass balance (BMB) from components as defined in Equation (2). All components are shown on a unified colorscale. Note that $\partial h/\partial t$ coverage was restricted to 81.5°S by orbital inclination prior to the 2010 launch of CryoSat-2 (Adusumilli et al., 2020)..*

B. Spatial validation scheme over Ross Ice Shelf

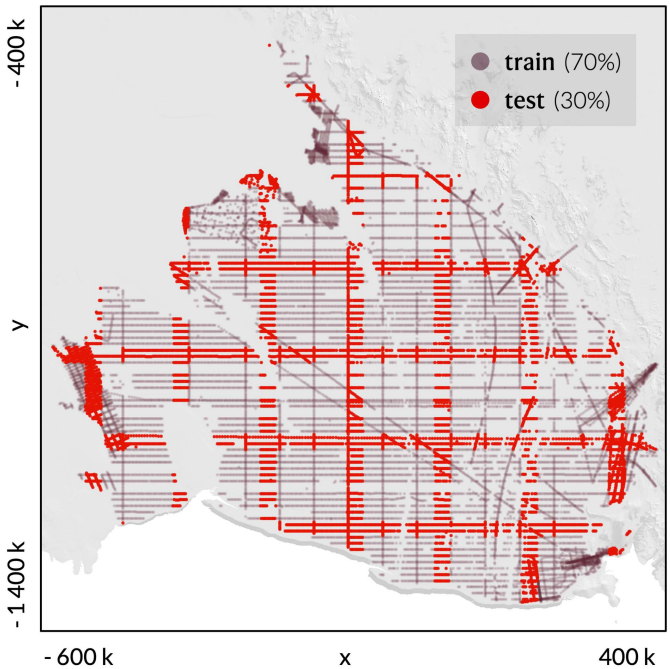

Figure 7. Train-test split over the Ross Ice Shelf. *The data is partitioned using a 'gingham' pattern to emulate the linear geometry of airborne flight tracks in this region. Test strips are 25 km wide, with 70% of ice flux observations (2.25 million) assigned to the training set and 30% (0.925 million) reserved for the test set.*

C. Hyperparameter ablations

We conduct additional ablation studies on the impact of the learning rate (lr) on the reconstruction quality of the two INR models, FluxNet (proposed) and ResMLP, in the experiments presented in Section 3.1 above. Ceteris paribus, we run the reconstruction experiments with $\text{lr} \in \{0.0005, 0.001, 0.002\}$, i.e. halving and doubling the learning rate relative to the default $\text{lr} = 0.001$, which was used in the experiments presented in the core paper. Here we perform 3 independent runs per learning rate and model to account for initialisation and training stochasticity. We present the ablation results in Table 2 and in Figure 8.

Table 2. *Test set predictive performance comparison of FluxNet and ResMLP on ice flux reconstruction over the Ross Ice Shelf for three different learning rates (lr) each. Metrics are indicated in units of ice flux, i.e. $m^2\ yr^{-1}$. Results represent the mean ± one standard deviation across 3 independent runs.*

	FluxNet (*proposed*)			ResMLP		
Metric	lr = 0.0005	lr = 0.001	lr = 0.002	lr = 0.0005	lr = 0.001	lr = 0.002
RMSE	9835 ± 582	9541 ± 562	53201 ± 71949	10972 ± 1201	11294 ± 834	10941 ± 735

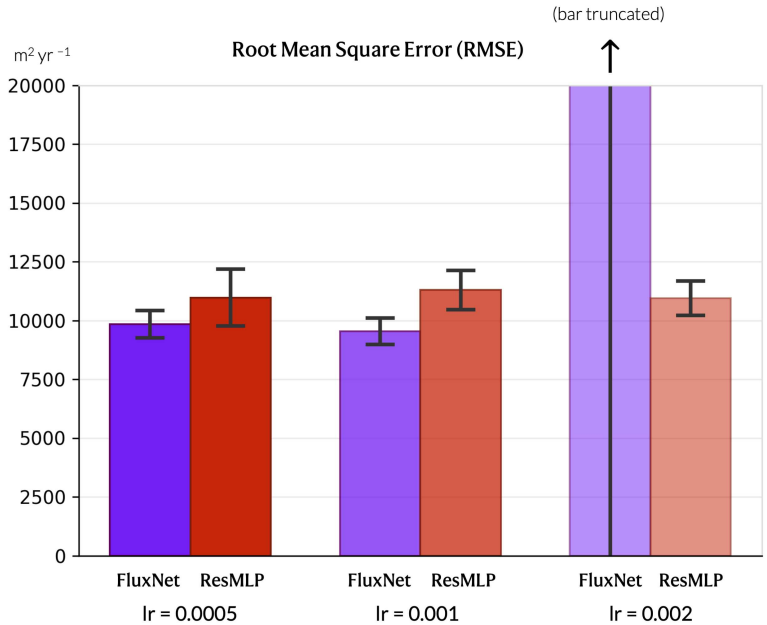

Figure 8. Ablation study bar chart corresponding to Table 2: *RMSE on the held-out test data for FluxNet and ResMLP, trained with three different learning rates (lr) each. The error bars represent* ± 1 *standard deviation calculated across the three experimental runs per model and lr. The result for FluxNet with lr = 0.002 is truncated to still show variations for results with smaller learning rates.*

This ablation study unveils that while ResMLP achieves consistent reconstruction skill across all three learning rates, it can not match the RMSE achieved by FluxNet at lower learning rates. FluxNet becomes unstable during training at higher learning rates, but overall is capable of higher performance. We hypothesise that this aptitude to lower learning rates in FluxNet is because the model outputs potentials ψ and ϕ rather than direct flux components q_x and q_y, and small changes in potentials may have large implications of the corresponding vector field. Overall this sensitivity to representation also allows FluxNet to learn physically-grounded nuance in target vector fields.

D. Overview of variables

Table 3. *Summary of variables and units used in the basal mass balance (BMB) formulation.*

Symbol	Description	Units
x, y	Horizontal spatial coordinates (e.g., polar stereographic)	m
BMB	basal mass balance rate (freeze-on > 0, melt < 0)	m yr^{-1} ice eq.
SMB	surface mass balance rate	m yr^{-1} ice eq.
h	ice thickness	m
t	Time	yr
$\frac{\partial h}{\partial t}$	Thickness rate of change	m yr^{-1}
$\mathbf{v} = (v_x, v_y)$	Depth-averaged horizontal ice velocity vector	m yr^{-1}
$\mathbf{q} = (q_x, q_y)$	Depth-integrated horizontal ice flux vector	m^2 yr^{-1}
$\nabla\cdot$	Divergence operator	—